



Jialan Zheng

Female, December 17th, 2002

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EDUCATION

09/2025 – present

Ph.D Student in Biomedical Engineering

School of Biomedical Engineering, Tsinghua University, China

- **Overall GPA:** 4.00/4.00, **Rank:** 1/55
- **Core Courses:** Principles of Magnetic Resonance Imaging (A+) / Physics in Nuclear Medicine Imaging (A+) / Big Data & Biostatistics (A) / Numerical Analysis (A) / Medical Imaging System (A-)

09/2021 – 06/2025

Bachelor of Science & Engineer (Dual Bachelor's Degree)

Chemical Biology & Biomedical Engineering, Tanwei College, Tsinghua University, China

- **Overall GPA:** 3.98/4.00, **Rank:** 1/15
- **Core Courses:** Calculus (A+) / Linear Algebra (A) / Chemical Principle (A) / Physics (A) / The Practice of C Programming (A+) / Introduction to Complex Analysis (A+) / Probability and Statistics (A+) / Electrical and Electronic Engineering (A) / Signals and Systems (A) / Biochemistry (A) / Structure and Function of Human Body (A) / Electrophysiological Principle and Experiment (A) / Digital Signal Processing (A-) / Medical Image Processing (A+) / Biomedical Electronics (A+) / Physiological System Simulation and Modeling (A+) / Biomedical Engineering Design Program (A+) / Basic Practical Biochemistry (A+) / Neural Modeling and Data Analysis (A+)

RESEARCH EXPERIENCES

4/2023 – present

Research related to diffusion MRI and neuroscience

Advisor: Qiyuan Tian, Department of Biomedical Engineering, Tsinghua University

◆ Project 1: Short-range Association Fiber (SAF) Mapping and Analysis:

- ✧ Acquired multi-resolution diffusion MRI data using submillimeter gSlider sequence; performed the first systematic, direct, and quantitative evaluation of SAF reconstruction across imaging and tractography strategies; and constructed the first human brain atlas of SAF proportions [J1, C1, C2];
- ✧ Quantified SAF myelin water fraction using the ViSta-MRF sequence [C3];
- ✧ Characterized spatial heterogeneity of superficial and deep white matter during early childhood white matter maturation [C4].

◆ Project 2: Generalizable Microstructure Quantification:

- ✧ Developed PIGMENT, a physics-driven foundation model that integrates large-scale, multicenter microstructural priors with subject-specific diffusion signal constraints to enable reliable quantification under limited acquisition constraints [C5, C6, M1];
- ✧ Developed MicroAgent, an autonomous agent framework that learns principles of self-supervised algorithm design, and enables zero-shot adaptive algorithm design for unseen acquisition conditions [M2].

◆ Project 3: MRI-Guided Precision Targeting for Neuromodulation

- ✧ Developed an automated gyrus-based mapping of the hand motor cortex from structural MRI, enabling rapid, individualized, and functionally informed VIM targeting for neuromodulation [C7, C8].

6/2024 – 9/2024

Diffusion MRI internship

Advisor: Susie Huang & Hong-Hsi Lee, Athinoula A. Martinos Center for Biomedical Imaging,

Massachusetts General Hospital, Harvard University

◆ Project: Estimating Fiber Diameter Using OGSE Sequence:

- ✧ Conduct OGSE scans on Connectome 2.0 scanner to estimate fiber diameter in a phantom [J2, C9, C10].

SELECTED JOURNAL PUBLICATIONS (#Co-First Author, *Corresponding Author)

- J1. **Jialan Zheng**, Ziyu Li*, Wen Zhong, Ziang Wang, Zihan Li, Hongjia Yang, Mingxuan Liu, Xiaozhi Cao, Congyu Liao, David H. Salat, Susie Y. Huang, Qiyuan Tian*. Effects of diffusion MRI spatial resolution on human brain short-range association fiber reconstruction and structural connectivity estimation. *Imaging Neuroscience* 2026; 4 IMAG.a.1089.
- J2. Dongsuk Sung*, Kwok-Shing Chan, Julianna Gerold, Wen Zhong, **Jialan Zheng**, Qiyuan Tian, Hua Guo, Susie Y Huang, Hong-Hsi Lee. Complementary sensitivity of fixed-time and fixed-oscillation regimes to exchange and structural disorder in the human brain revealed using oscillating-gradient diffusion MRI with ultra-strong gradients. *Magn Reson Med.* 2026 Jun; 95(6):3429-3444.

SELECTED CONFERENCE PUBLICATIONS

- C1 **Jialan Zheng**, Cong Yang, Wen Zhong, Ziang Wang, Haoxiang Li, Zihan Li, Mingxuan Liu, Hongjia Yang, Xiaozhi Cao, Congyu Liao, David H. Salat, Susie Y. Huang, Qiyuan Tian. Effect of Diffusion MRI Acquisition and Nominal Spatial Resolutions on Fiber Reconstruction and Connectivity Estimation. *ISMRM Workshop on 40 Years of Diffusion*. 2025. (Oral)
- C2 **Jialan Zheng**, Cong Yang, Wen Zhong, Ziang Wang, Haoxiang Li, Zihan Li, Mingxuan Liu, Hongjia Yang, Xiaozhi Cao, Congyu Liao, David H. Salat, Susie Y. Huang, Qiyuan Tian. Impacts of diffusion MRI spatial resolution on the short-range association fiber reconstruction and connectivity estimation. *The Annual Meeting of International Society of Magnetic Resonance for Medicine (ISMRM)*. 2025. (Poster)
- C3 **Jialan Zheng**, Cong Yang, Wen Zhong, Ziang Wang, Haoxiang Li, Zihan Li, Mingxuan Liu, Hongjia Yang, Xiaozhi Cao, Congyu Liao, David H. Salat, Susie Y. Huang, Qiyuan Tian. Quantification of Human Brain White Matter Myelination using ViSTa MRI Fingerprinting. *The Annual Meeting of Organization of Human Brain Mapping (OHBM)*, 2025 (Poster).
- C4 **Jialan Zheng**, Zihan Li, Xun Yuan, Hongjia Yang, Yi Liao, Haibo Qu, Qiyuan Tian. Tractography-Based Mapping of Spatial Patterns in Early Childhood White Matter Maturation. *The Annual Meeting of Organization of Human Brain Mapping (OHBM)*. 2026. (Poster)
- C5 Zihan Li, **Jialan Zheng**, Ziyu Li, Ziang Wang, Mingxuan Liu, Guochen Ning, Hongen Liao, Qiyuan Tian. DIMOND++: Improving diffusion model optimization using diffusion priors. *The Annual Meeting of International Society of Magnetic Resonance in Medicine (ISMRM)*, 2025. (Oral Power Pitch) [MERIT SILVER AWARD]
- C6 Zihan Li, **Jialan Zheng**, Ziyu Li, Zhuohao Wang, Mingxuan Liu, Hongjia Yang, Guochen Ning, Hongen Liao, Qiyuan Tian. DIMOND++: Improving diffusion model optimization using diffusion priors. *ISMRM Workshop on 40 years of Diffusion: Past, Present & Future Perspectives*, 2025. (Poster)
- C7 Ziang Wang, **Jialan Zheng**, Zihan Li, Hongjia Yang, Jennifer McNab, Zuoxiang He, Qiyuan Tian. Functional Hand Motor Cortex Is Not Always on Anatomical "Handknob". *The Annual Meeting of Radiology Society of North America (RSNA)*. 2025. (Poster)
- C8 Ziang Wang, **Jialan Zheng**, Kasidit Anmahapong, Hongjia Yang, Zuo-Xiang He, Xue Zhang, Qiyuan Tian. Beyond the Hand Knob: Automated Gyrus-Based Mapping of the Hand Motor Cortex for Precise VIM Targeting. *The Annual Meeting of International Society of Magnetic Resonance for Medicine (ISMRM)*. 2026. (Poster)
- C9 **Jialan Zheng**, Wen Zhong, Dongsuk Song, Julianna Gerold, Qiyuan Tian, Hua Guo, Susie Y. Huang, Hong-Hsi Lee. Revealing Diffusion Frequency-Dependence in Surface-to-Volume Ratio limit using OGSE sequence on the Connectome 2.0 Scanner. *The Annual Meeting of International Society of Magnetic Resonance in Medicine (ISMRM)*, 2025 (Poster).
- C10 Dongsuk Sung, Kwok-Shing Chan, Julianna Gerold, Wen Zhong, **Jialan Zheng**, Qiyuan Tian, Hua Guo, Susie Y Huang, Hong-Hsi Lee. Probing water exchange and structural disorder in the human brain with oscillating-gradient diffusion MRI. *The Annual Meeting of International Society of Magnetic Resonance for Medicine (ISMRM)*. 2026. (Oral)
- C11 **Jialan Zheng**, Jiachen Liu, Cong Yang, Hongjia Yang, Zihan Li, Haoxiang Li, Haiyan Ding, Xihai Zhao, Qiyuan Tian. Short-term White Matter Microstructural Changes in Young Adults Infected with SARS-COV-2 Omicron Variant. *The Annual Meeting of International Society of Magnetic Resonance in Medicine (ISMRM)*, 2024. (Poster)
- C12 Zihan Li, **Jialan Zheng**, Ziyu Li, Yi Jing, Hongen Liao, Qiyuan Tian. Improving Microstructure Quantification using Self-supervised and Physicsinformed Deep Learning. *The Scientific Assembly and Annual Meeting of Radiological Society of North America (RSNA)*, 2024 (Poster)

MANUSCRIPTS UNDER REVIEW (#Co-First Author, *Corresponding Author)

- M1. Zihan Li[#], **Jialan Zheng**[#], Ziyu Li^{*}, Xun Yuan, Kasidit Anmahapong, Ziang Wang, Mingxuan Liu, Hongjia Yang, Yifei Chen Zhuohao Wang, Yuhang He, Fang Chen, Rui Li, Huaiqiang Sun, Yi Liao, Congyu Liao, Yang Yang, Haibo Qu, Xue Zhang, Qiyuan Fan, Susie Y. Huang, Hongen Liao^{*}, Qiyuan Tian^{*}. A physics-informed foundation model enables quantitative microstructure mapping from sparse and heterogeneous diffusion MRI. *Science Bulletin*. 2026. (Under review)
- M2. **Jialan Zheng**[#], Zihan Li^{#*}, Xun Yuan, Mingxuan Liu, Ziang Wang, Kasidit Anmahapong, Hongjia Yang, Yifei Chen, Rui Li, Yi Liao, Haibo Qu, Ziyu Li^{*}, Qiyuan Tian^{*}. MicroAgent: Agentic learning for microstructure quantification from diffusion MRI. *IEEE Transactions on Medical Imaging*. 2026. (Under review)

HONORS AND AWARDS

2025	Scholarship for Comprehensive Excellence, Tsinghua University
2025	Outstanding Bachelor's Graduate of Beijing.
2025	Outstanding Bachelor's Graduate of Tsinghua University.
2025	Future Scholars Scholarship.
2024	National Scholarship, Ministry of Education, P.R. China
2023	National Scholarship, Ministry of Education, P.R. China
2023	Scholarship for Science and Technology Innovation Excellence, Tsinghua University
2022	Scholarship for Comprehensive Excellence, Tsinghua University
2022	Scholarship for Academic Excellence, Tanwei College

FUNDING

- F1. **Jialan Zheng**, Xuguang Bai. Diffusion MRI Based Study of Human Brain Development. *Qi Yan Undergraduate Research Project*, Beijing Municipal NSF, 2024. (¥50K)
- F2. **Jialan Zheng**, Haoxiang Li, Mingxuan Liu. Diffusion MRI Based Study of Human Brain Development. *Initiative Scientific Research Program*, Tsinghua University, 2024. (¥19K)
- F3. **Jialan Zheng**. High-Resolution Diffusion Magnetic Resonance Imaging Study of Brain Structural Connectivity. *Student Research Training (SRT) Program*, Tsinghua University, 2024. (¥5K)

SKILLS

Programming and Tools: Python, PyTorch, MATLAB, C, Linux, Verilog, STM32, LaTeX

Language: Mandarin Chinese (native), English (TOEFL: 93)